

Shivam Kumar Verma

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[LinkedIn](#)

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[Portfolio](#)

Education

Indian Institute of Information Technology Nagpur

Nagpur, Maharashtra

B.Tech in Computer Science and Engineering | CGPA: 7.5

Nov 2022 – May 2026

Relevant Coursework: Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Skills

Languages: TypeScript, JavaScript, Python, C++ (competitive programming)

Frameworks: Node.js, Express.js, Next.js, React, FastAPI

Databases & ORMs: PostgreSQL, Redis, ClickHouse, Prisma, Drizzle

DevOps & Cloud: Docker, AWS (EC2, S3, Lambda), Nginx, Linux, Git, CI/CD

AI & ML: LLM integration, Prompt engineering, Semantic search, Pinecone

Experience

Software Developer Intern – klimb.io

Nov 2025 – Current

TypeScript · React · Node.js · Express.js · PostgreSQL · Redis · AWS (S3, EC2)

- **Credit-Based Monetization:** Redesigned platform access control by migrating from feature flags to a credit-based usage system using PostgreSQL and middleware guards, enabling monetization of **6+ AI features** and cutting complexity by **~70%**.
- **Global Field Template Architecture:** Replaced per-tenant field duplication with a shared global template model by restructuring the DB schema and backfilling data, achieving **zero-downtime migration** with **complete data integrity** across all customers.
- **Dashboard Latency Optimization:** Eliminated expensive runtime aggregations by introducing precomputed, indexed data pipelines, slashing recruiter dashboard load time from **2–3s to under 500ms**.

Projects

CodeSM

[Live Link](#)

React · Node.js · Express.js · Redis · BullMQ · Docker · PostgreSQL · AWS EC2

- **Scalable RCE Engine:** Built an async code execution engine by implementing segregated BullMQ queues and lightweight Redis payload caching, enabling reliable high-throughput submissions with **near-zero dropped jobs**.
- **Persistent Container Strategy:** Eliminated container spin-up overhead by switching to persistent Docker containers with tmpfs mounts, reducing per-test execution time from **300ms to 5ms (~98% faster)**.
- **Fault-Tolerant Worker Architecture:** Hardened job processing by adding a **Dead Letter Queue** for retries and a 1-hour Redis cache layer, significantly reducing DB load during **high-traffic spikes**.

Storix

[Live Link](#)

Next.js · TypeScript · FastAPI · Celery · Redis · PostgreSQL · AWS S3

- **Async PDF Pipeline:** Decoupled PDF ingestion from processing by introducing distributed Celery workers and a Redis task queue, enabling non-blocking long-running jobs with **no impact on API response times**.
- **Direct Cloud Uploads:** Offloaded file transfer from the server by generating pre-signed S3 URLs for direct client uploads, **eliminating backend bottlenecks** and improving upload reliability.
- **Real-Time Job Status via WebSockets:** Replaced polling with a Pub/Sub and WebSocket layer to push live task updates, giving users **instant feedback** from upload through delivery.

Chatterly

[Live Link](#)

Next.js · Socket.IO · Redis · PostgreSQL · Prisma · NextAuth

- **Horizontally Scalable Messaging:** Enabled multi-instance chat by wiring Redis Pub/Sub for cross-server event propagation, ensuring **consistent room state** across all nodes **without sticky sessions**.
- **Zero-Drop Room Event Routing:** Isolated room traffic by subscribing each server to dedicated Redis channels, guaranteeing **zero message drops** under concurrent multi-user load across distributed instances.
- **JWT Auth & Room Authorization:** Protected private rooms by enforcing JWT verification and room-level checks at the Socket.IO middleware layer, **fully preventing unauthorized access**.

Achievements

- **Competitive Programming:** Ranked **top 5% globally** on [LeetCode](#) (**1,800+ rating**) · **3-star** on [CodeChef](#) (1,700+ rating) · Pupil on [Codeforces](#) (max 1,261) · Rated on [AtCoder](#) (440+)